

Input

Model

Shapes and training

Problem

Fix

Text to numbers
integer encoding,
padding
Note 1057

Word embeddings
Note 1057

**Recurrent
neural network**
reads one item at a time
Note 1055–1056

Same weights at
every time step
Note 1056

Types of RNN
many-to-one,
one-to-many,
many-to-many
Note 1058

**Backpropagation
through time**
Note 1059

Backpropagation
Note 1015–1017

Vanishing gradient
Note 1018

**Long-term
dependency**
far inputs are forgotten
Note 1060

Unstable training
exploding gradient
Note 1060

LSTM and GRU
Note 1061–1066



needs



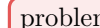
is a kind of (general to specific)



fixes



used in



problem



from another map